

Balancing Point

The **balancing point** of a dataset is the point on a number line where the data distribution is balanced.

1. Use the instructions below to find the balancing point of the following set of numbers:
2, 3, 6, 8, 9, 11.

Instructions:

Step 1: Hover over a number on the ruler and click to add a penny at that position.

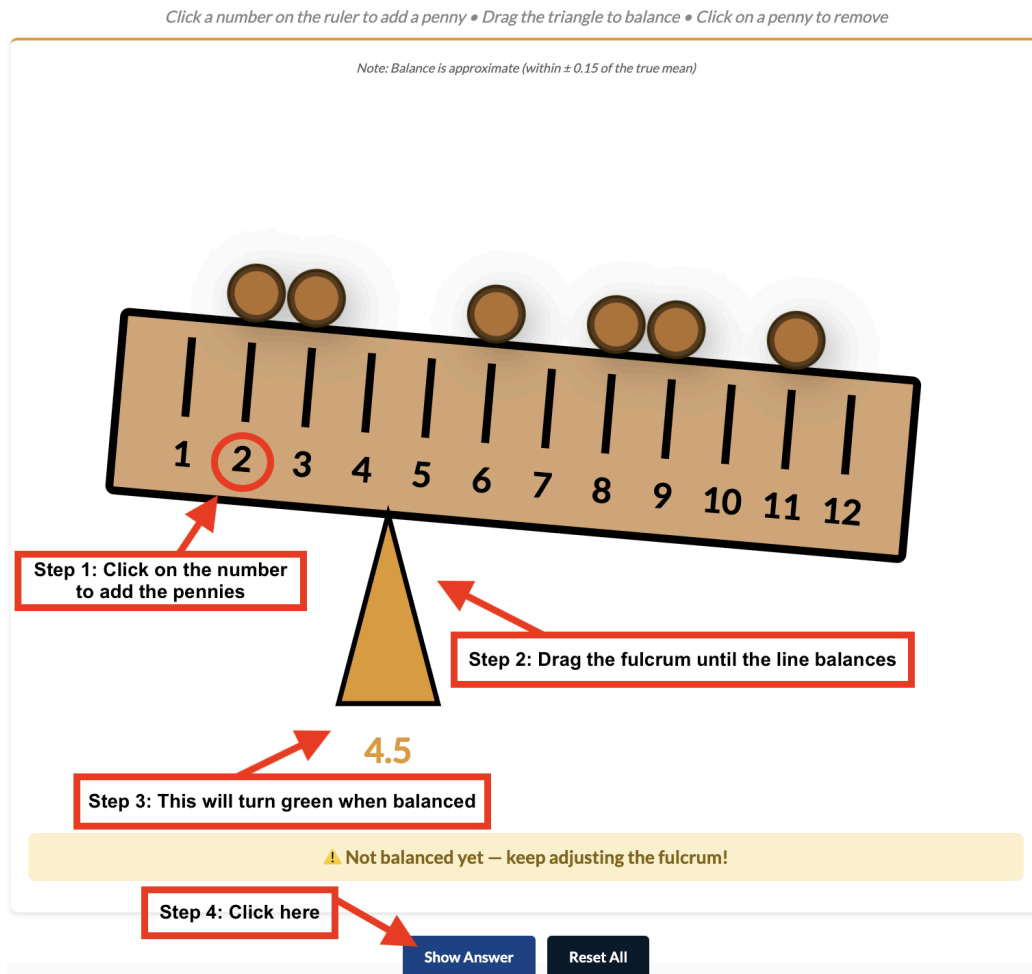
Step 2: Drag the fulcrum (triangle) left or right to balance the ruler.

Step 3: When balanced, the ruler will be level and turn green.

- a. To the nearest half inch, at what value on the ruler is the data balanced?

Step 4: Click "Show Answer" to have the computer calculate the exact balancing point (mean).

- a. Now, find the actual mean of the dataset. What do you notice?



Name_____

Date_____

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2. Answer the following questions:

a. Use the balancing method (step 3 from above) to find the mean of each dataset below:

i. 2, 2, 8, 9, 9

ii. 1, 3, 4, 7, 8, 10

iii. 4, 5, 5, 9, 11, 11

b. Suppose a line with several tokens is balanced. What happens when you move some of the tokens to the right? To the left? Explain how this affects the mean.

c. What does the balancing point of a dataset represent? Explain.